Supplementary Information

Learning conditional language in continuous image space

Imagized Language Model Project

Supplementary Note 1: experimental question and scope

The target is an independent visual language model whose runtime consumes writing images and generates writing images. Encoded text and local language models may prepare data offline. They cannot supply the answer during model inference. The present experiment isolates canonical Chinese continuation as one prerequisite. It does not cover the complete instruction-following or historical-script objective.

The repository retains the earlier 96-page research record in its legacy image-native publication folder. That record documents prior experiments without treating every rejected architecture as a contribution to the compact article. This supplement specifies the directly conditional block test, the subsequent exact-raster density and the distinction between ranking and generation.

Supplementary Note 2: conditional uncertainty

Let H denote a visible history and X the next writing image. Under squared pixel loss, the optimum $f^*(H) = \mathbb{E}[X \mid H]$ can mix valid glyphs. Under independent Bernoulli loss, each marginal equals $\Pr(X_p = 1 \mid H)$, which likewise does not specify joint stroke identity. An image distribution is consequently required for stochastic writing.

Selection and sampling have different expected identity accuracy. If a model exactly recovers the true next-form distribution $p(x \mid H)$, deterministic top-1 identity has conditional accuracy $\max_x p(x \mid H)$. Independently sampling one future and comparing it to one observed continuation has expected accuracy $\sum_x p(x \mid H)^2$. Neither quantity is a general measure of semantic correctness. Thus the reader's 19.97% top-1 cannot be used as the expected accuracy of a correct stochastic writer. Paired context controls, sampled probabilities, direct readability and meaningful longer continuations are needed together.

Supplementary Note 3: raster transform and gradients

For signed ink image $S = 2X - 1$ and a 32×32 orthonormal basis B, the visual coordinate is

$$\phi(S) = \frac{\operatorname{vec}(BSB^{\mathsf{T}})}{\|\operatorname{vec}(BSB^{\mathsf{T}})\|_2}. \tag{1}$$

For binary observed images this matches the reader's field. For soft predicted images it must be computed directly on S. Applying the reader's input threshold inside this training objective would cut all pixel gradients. The pretraining inspection found and corrected that defect. A regression test confirms a finite, nonzero gradient from the field cosine alone to the predicted pixels and equality with the reader transform on binary inputs.

The transform is a full spatial coordinate, not a learned visual vocabulary. Its background contribution also limits its interpretation as a perceptual identity metric. A high cosine can accompany an incorrect glyph.

Supplementary Note 4: directly conditional architecture

The frozen reader has 16,277,249 parameters. Its final 384-dimensional hidden state and 1,024-dimensional anchor form a continuous condition. A 5,899,970- parameter U-Net receives a complete 32×128 future strip. Three spatial scales have 64, 128 and 256 channels. Each residual scale receives the projected condition and a sinusoidal time embedding. Additional input channels carry the first-cell predicted plan, a horizontal coordinate and the cell noise levels.

At the bottleneck, four pooled cell vectors enter four-head attention and a 512-wide MLP. A zero-initialized scalar gates the spatial mixer residual. The output channel predicts velocity over four visible cells. The total runtime contains 22,177,219 parameters. The prior checkpoint only initializes matching U-Net weights. An epsilon predictor is not thereby a qualified velocity predictor, so all copied layers are fine-tuned under the new loss.

Supplementary Note 5: data and optimization

The corpus manifest SHA-256 is

```
76048753b52735d14c98f0d9e4eb8d751401fe8810e82eaaaebff517f6866c03.
```

Source identifiers define deterministic splits before rendering. Original and OpenCC-simplified views inherit the same split. Data generation samples 65-cell segments with a saved dataset seed. Only the first 64 cells are used in this test. A training example has 60 context and four target cells. A symbolic character identifier is never an input or target to a model method.

On odd updates, the four future positions receive one level from each of 0–24, 25–59, 60–89 and 90–99 under a random permutation. On even updates, one uniformly sampled level from 0–99 applies to all four future cells. The latter pretraining amendment supplies synchronous noise patterns like those encountered during sampling. It was frozen before optimization.

Ten percent of rows drop the condition. All-dropped batches have zero order loss. The foreground loss weights pixels by $1+2X$. The complete objective is

$$L = L_v + 0.10L_{fg} + 0.10L_{dice} + 0.10L_{field} + 0.10L_{repr} + 0.25L_{order}, \tag{2}$$

$$L_{order} = \mathbb{E}_{t\geq 60}\,\mathrm{softplus}\left[0.02 - (c_{ordered} - c_{shuffled})\right]. \tag{3}$$

Only rows with an available condition enter the order loss. The reader remains frozen. Noise, drop masks and data offsets derive from saved RNG states. The run uses seed 20265310, dataset seed 20265311, AdamW with $\beta = (0.9, 0.95)$, weight decay 0.01, gradient norm clip 1, EMA decay 0.999, batch eight and 8,000 updates. The full run consumes 64,000 context segments and 256,000 supervised future cells. These are sampled segments rather than 64,000 independent source records.

Supplementary Note 6: locked audit and uncertainty

The audit seed is 20265320. It uses 512 fixed development windows for denoising, 256 for direct generation and 64 for two-block visible feedback. The first 60 cells form the context and the remaining four context-window cells form the first target block. The four subsequent continuation cells form the second block. Denoising probes use levels 75 and 95, with identical noise under all conditions. Prefix shuffling preserves the final four visible cells. The zero condition removes all reader state and plan channels.

The 1,024-image bank is rendered after every denoising and generation call. Every primary output is the first sample. Nearest-bank identity is evaluator metadata, not an image replacement or glyph selected by the model. Per-position coverage records how many true targets appear in the bank. All out-of-bank targets receive zero exact-identity credit. The rendered output remains the actual model image even when a bank identity is assigned.

Bootstrap uncertainty resamples source identifiers, retaining all correlated windows from a record together, with 2,000 resamples. Paired differences keep the same starting noise. The intervals do not include training-seed variation. There is one primary trained seed and no sealed generalization result.

Supplementary Table 1: direct visual block audit

Condition	First identity (%)	Block identity (%)	Pixel F1	Isolated ink
Ordered	1.172	0.586	0.3198	0.0050
Shuffled	1.172	0.488	0.3176	0.0053
Zero condition	0.000	0.195	0.3018	0.0053

All 256 prefixes contribute one four-cell sample. Target coverage in the Han bank is 60.94, 58.98, 64.45 and 68.36% by position. No out-of-bank target receives identity credit.

Supplementary Note 7: numerical and compute receipts

Training takes 361.6 seconds and peaks at 0.543 GiB of allocated CUDA memory on GPU 0, an RTX 4090 D. These figures describe the bounded experiment. They exclude prior foundation training, data preparation, idle GPU memory, operating-system memory and evaluation. They are not an estimate of the cost of a broadly capable language model. The independent reader and prior training costs remain in their original run reports.

The continuous versus interrupted four-step smoke check compares every denoiser and EMA tensor and the saved training generator. All match exactly. Production checkpoints save optimizer groups, CPU/CUDA/Python/NumPy RNGs and the deterministic data offset. Foundation, data, source and protocol hashes are checked before scoring. No other project's GPU service is part of this run.

Supplementary Note 8: limitations and next evidence

The post-result motor diagnostic uses the first 64 fixed prefixes and paired noise. Replacing the first plan with the exact target raster gives F1 0.4586. Replacing both plan and visual anchor gives F1 0.4594 and identity 5/64, compared with F1 0.3775 and identity 1/64 for the predicted condition. An exact inverse-transform control has F1 1.0. Only 60.94% of these targets occur in the Han identity bank, so its 60.94% identity is the available ceiling for perfect copies. The target-supplied test was designed after the locked audit and does not change its decision. It isolates a remaining motor limitation.

Reliable autonomous continuation, calibrated conditional sequence likelihood, held-out short answers, font transfer, independent seed replication and capability-matched text baselines remain outstanding. No selected gallery can substitute for these tests. Future teacher-generated exercises must retain source provenance and separate prompt/answer splits. A deployed teacher or an image-bank lookup would change the model contract.

The older guided model's terminal-signal observation is a mathematical limit, not a proven failure explanation. Correct conditional diffusion also starts from a context-independent noise law. Direct interventions on trained conditional scores and matched initialization are needed to distinguish weak condition learning from a defective sampling trajectory.

Supplementary Note 9: normalized raster density

The exact-density model (V54) freezes the same 16,277,249-parameter reader. A row transformer has width 192, four layers, six heads, MLP ratio three and zero dropout. Its inputs are shifted earlier rows, a projection of the aligned 32-pixel plan row, spatial row positions and a projected continuous condition (hidden384 plus field1024). A MADE with two 256-wide SiLU layers masks pixels within each row. Its output mask is strict: pixel c cannot see pixel c or any later pixel. Earlier rows may be observed in full. Gradients with respect to forbidden pixels vanish even with nonzero randomized output weights. Enumeration of all 16 four-bit outcomes gives total probability one.

The inverse reader field, scaled by 32 and clipped to $[-2,2]$, supplies an aligned logit at scale six. Residual output weights start at zero. Thus a supplied binary image has greedy reconstruction exactly equal to itself at initialization. This copy path does not learn language and is not a lookup bank. It applies to arbitrary supplied binary rasters. The experiment measures whether optimization retains that path while learning natural continuation.

Each update samples eight streams and 16 causal positions per stream, four from each quarter, including the last position. Of the resulting 128 target images, 25% in expectation receive their actual image field and a zero hidden state as explicit motor exercises. The remainder receive frozen predicted states. Antialiased target images are thresholded at 0.5 before unweighted Bernoulli NLL; soft labels are rejected by the loss. There is no additional foreground, contrastive or guidance objective. Seeds are 20265410 for training and 20265411 for data. AdamW uses learning rate 2×10^{-4}, $\beta=(0.9,0.95)$, weight decay 0.01, 400 warmup updates, cosine decay to 10%, norm clip one, BF16 and EMA 0.999. The run makes 1,024,000 selected-image presentations over 8,000 updates; these are not independent records.

Training completes in 207.8 s with 0.400 GiB peak allocated CUDA memory. The runtime has 18,711,841 parameters, of which 2,434,592 are new density parameters. The 52.16-second audit peaks at 0.231 GiB allocated CUDA memory. Neither timing includes earlier reader training. Source hashes, the corpus hash, optimizer, all RNGs and the dataset offset are saved. An interrupted-versus-continuous smoke run matches every model and EMA tensor and the training RNG exactly. The production checkpoint SHA-256 is

`f2c3fab74981541243a963c8cffab6bbbd8c7f9d051512b228347db04a57a9ee.`

Supplementary Note 10: exact-density audit and decision

Audit seed 20265420 selects 512 development windows from 169 source records for exact target NLL. Pixel autoregression is teacher forced only for this likelihood calculation. The paired earlier-history shuffle preserves the last four visible cells. Ordered NLL is 15.219 nats per image and shuffled NLL is 15.254. Their advantage is 0.0355, with interval [-0.1014, 0.1693] from 2,000 source-record draws (bootstrap seed 20265519). The zero-condition NLL is 47.930 nats; zeroing is an out-of-distribution control, not an optimized unconditional model.

Generation uses 256 prefixes with paired Bernoulli uniforms. Within each image, 32 row-transformer and 1,024 masked-head evaluations produce pixels without rejection or reranking. All generated images precede evaluator-bank construction. The bank combines all distinct training-form rasters with every target raster from generation, motor and feedback tests. Pixel deduplication gives 9,932 entries and full target coverage, including punctuation. Its nearest identity remains a proxy and can credit a malformed image. It differs from the 1,024-Han bank used in earlier experiments.

Supplementary Table 2: exact raster density audit

Condition	Identity (%)	Exact image (%)	Pixel F1	Ink ratio
Ordered sample	1.562	0.781	0.35033	0.945
Shuffled sample	1.172	0.391	0.35124	0.945
Zero-condition sample	0.000	0.000	0.19715	0.435
Ordered greedy	5.469	5.469	0.21454	0.330
Supplied image, sample	100.000	81.250	0.99949	1.000
Supplied image, greedy	100.000	100.000	1.00000	1.000
Deranged image, greedy	0.391	0.391	0.35769	1.000

Each row uses 256 images. The evaluator bank contains 9,932 distinct rasters and covers every target. Supplied-image rows reveal the intended image and are motor controls, not language results.

The paired ordered-minus-shuffled identity difference is 0.391 percentage points with interval [0, 1.210], using 134 source records. Four-step visible feedback on 64 prefixes reaches mean identity 4.297% versus 3.125% shuffled. Ordered step-wise identity is 4.688, 4.688, 4.688 and 3.125%; shuffled values are 4.688, 1.563, 3.125 and 3.125%. These secondary measurements do not repair the failed primary context effect.

Four of eight preregistered gates pass: runtime boundary, motor copying, positive feedback difference, and finite/resource checks. Ordered likelihood, ordered sampled identity, sampled quality and isolated-ink surface gates fail. The primary sample has isolated-ink fraction 0.01404, above the 0.01 limit. The decision remains `reject`. No sealed evaluation or independent training-seed replication is reported.

Supplementary Note 11: independent inference export

An inference-only checkpoint embeds the reader, row model, masks and configuration. It loads without original foundation checkpoints. An aligned 32-pixel image grid is the native prompt interface; optional host text rendering does not pass symbolic identities into the network. Eight-cell greedy output from the exported checkpoint is pixel-identical to the training checkpoint's output with the same prompt. The native motor CLI also matches the binarized input exactly. The export is a 75,225,847-byte research artifact, not a qualified conversational model. General instruction following, open-script language generation and etymology remain unestablished.

Supplementary Note 12: exploratory context attribution

A plan written after the locked result specifies a read-only likelihood diagnostic on the same 512 windows. It leaves the checkpoint, primary audit and rejection unchanged. Ordered and earlier-shuffled NLL reproduce the saved per-example values within 7.63×10^{-6} nat. To derange the complete condition, windows are sorted by source record and the index list is shifted by half its length. Every donor source differs from its target source. Both hidden state and predicted field come from that donor. No image bank or optimization enters this intervention.

Complete-condition derangement raises NLL from 15.219 to 20.630 nats/image. The paired ordered advantage is 5.411 nats, with source-record bootstrap 95% interval [4.514, 6.416]. Earlier-history shuffling, which retains the final four cells, still gives the unresolved 0.0355-nat difference. The density therefore uses the visual condition; its earlier-history ordering effect remains weak. Short-suffix or order-insensitive features are compatible explanations, not an established exclusive mechanism. The 512-image answer-supplied NLL is 0.190 nat; deranging that supplied image raises NLL to 1,954.872 nats. These answer-revealing controls describe the motor path.

The first eight image rows contain 8.644 of 15.219 ordered nats. Whole-context derangement adds 2.365 nats in that band and 1.530, 1.127 and 0.390 in the remaining eight-row bands (Fig. S1). Earlier-shuffle effects vary in sign across bands. No band-wise significance claim or multiple-comparison-adjusted result is made. These model-specific likelihood interventions are not calibrated mutual-information estimates. They do not establish autonomous coherent writing. The next mechanism must retain this context sensitivity while selecting globally coherent future forms.

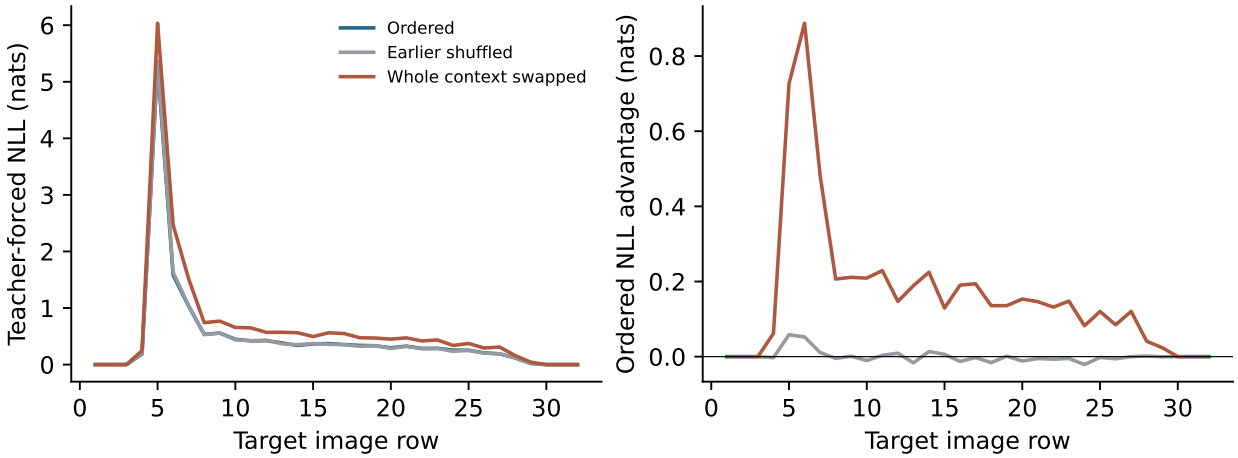


Fig. S1 | Exploratory row-wise attribution of raster likelihood. Left, teacher-forced NLL by target-image row for ordered, earlier-shuffled and completely deranged conditions on 512 windows. Right, ordered advantage relative to each control. Earlier target pixels are legitimately observed by the image density, so low later-row NLL cannot itself establish language information from the prompt. Curves are descriptive means; whole-image uncertainty is reported in Supplementary Note 12.

Supplementary Note 13: matched joint likelihood learning

The joint experiment (V55) retains the exact V54 network and probability law. Both arms initialize from its EMA, with new optimizers and EMA copies. The frozen control trains only the 2,434,592 density parameters. The joint candidate also trains reader parameters through the exact image likelihood, for 18,711,840 participating parameters. The unused scalar contrastive temperature is fixed. Runtime size stays 18,711,841 in both. Reader dropout remains disabled in both arms, independently of gradient flow. The deployed student continues to receive and generate image tensors, without a vocabulary or external model.

Each arm makes 8,000 updates with eight streams and 16 selected causal positions per stream, using 25% supplied-image motor exercises. Training seed 20265510 and dataset seed 20265511 give matched new draws from the same public corpus. Both final training-generator states match exactly. The frozen reader is bit-identical to initialization; the joint reader changes. Reader likelihood gradients are finite and nonzero only in the joint arm. A four-step interrupted smoke reproduces every model and EMA tensor, plus CPU/CUDA/training RNG states. Reader gradients cannot pass through answer-supplied motor rows because those replace the natural state with a zero hidden vector and a fixed observed field.

The density learning rate is 2×10^{-4} in both arms; joint reader rate is 2×10^{-5}. Other optimizer, warmup, cosine, clipping, BF16 and EMA settings match the earlier density experiment. The control takes 218.0 s at 0.400 GiB allocated CUDA memory; the joint arm takes 342.4 s at 0.648 GiB. Both run sequentially on physical GPU 0. These costs exclude V42/V54 pretraining and data preparation.

Both checkpoints finish before either development result opens. The same 512 NLL windows, 256 generation/motor prefixes and 64 four-step feedback streams use audit seed 20265420. Joint ordered NLL is 7.026 nats per image versus 7.579 for the matched frozen control. The paired gain is 0.553 nat, interval [0.450, 0.665]. The joint-minus-control sample identity gain is 1.172 percentage points, interval [-0.397, 2.893]. All intervals use 2,000 source-record bootstrap draws. Joint earlier-history advantage is 0.185 nat [0.128, 0.244], versus 0.119 [0.052, 0.186] for the continued frozen arm. Extra training accounts for part of the likelihood progress; the matched gain isolates the reader update's additional benefit.

Supplementary Table 3: matched learning audit

Arm / decoding	Identity (%)	Exact image (%)	Pixel F1	Isolated ink
Frozen / ordered	2.344	1.953	0.3643	0.0070
Frozen / shuffled	2.344	1.953	0.3638	0.0071
Frozen / greedy	6.641	6.641	0.2296	0.0028
Joint / ordered	3.516	3.516	0.3620	0.0074
Joint / shuffled	3.125	3.125	0.3625	0.0065
Joint / greedy	7.422	7.422	0.2230	0.0024

Both arms use 256 paired prefixes and the same post-generation evaluator bank. No sampled output is selected or replaced.

Both arms preserve exact greedy copying on 256/256 targets. Stochastic-copy F1 is 0.99993 frozen and 0.99994 joint; exact-image match is 96.875% and 97.266%, respectively. Joint four-step feedback identity is 3.125% ordered and 1.563% shuffled; control values are 2.344% and 2.734%. The joint arm passes six of eight unchanged gates, failing sampled quality and the ordered generation advantage. The control passes five, additionally failing feedback. The matched decision remains `reject`. Neither the primary development reuse nor the paired intervals establish independent seed or sealed-partition generalization.

Supplementary Note 14: form validity after joint learning

A post-result diagnostic compares saved images with the same 9,932 distinct canonical rasters used for evaluation. Exact membership asks whether an output is a known form; exact target match asks whether it equals the observed next image. Neither uses nearest-neighbour replacement. The bank is confined to evaluation and never enters decoding. These descriptive measurements reuse development outputs and do not change any original decision.

Among 256 first samples, exact known-form membership rises from 9.375% in V54 to 33.594% in the continued frozen arm and 43.750% in the joint arm. The paired joint-minus-frozen gain is 10.156 percentage points, with 95% source-record interval [4.219, 16.240]. Exact observed-target matches are 0.781, 1.953 and 3.516%, respectively. In four-step feedback, joint form membership is 46.875% versus 38.672% frozen; the paired gain is 8.203 points [1.230, 14.683]. The first-sample joint greedy output has 98.438% membership but only 7.422% exact target match and 34 distinct forms, compared with 78 distinct stochastic outputs. Form coherence and linguistic choice remain separate requirements (Fig. S3).

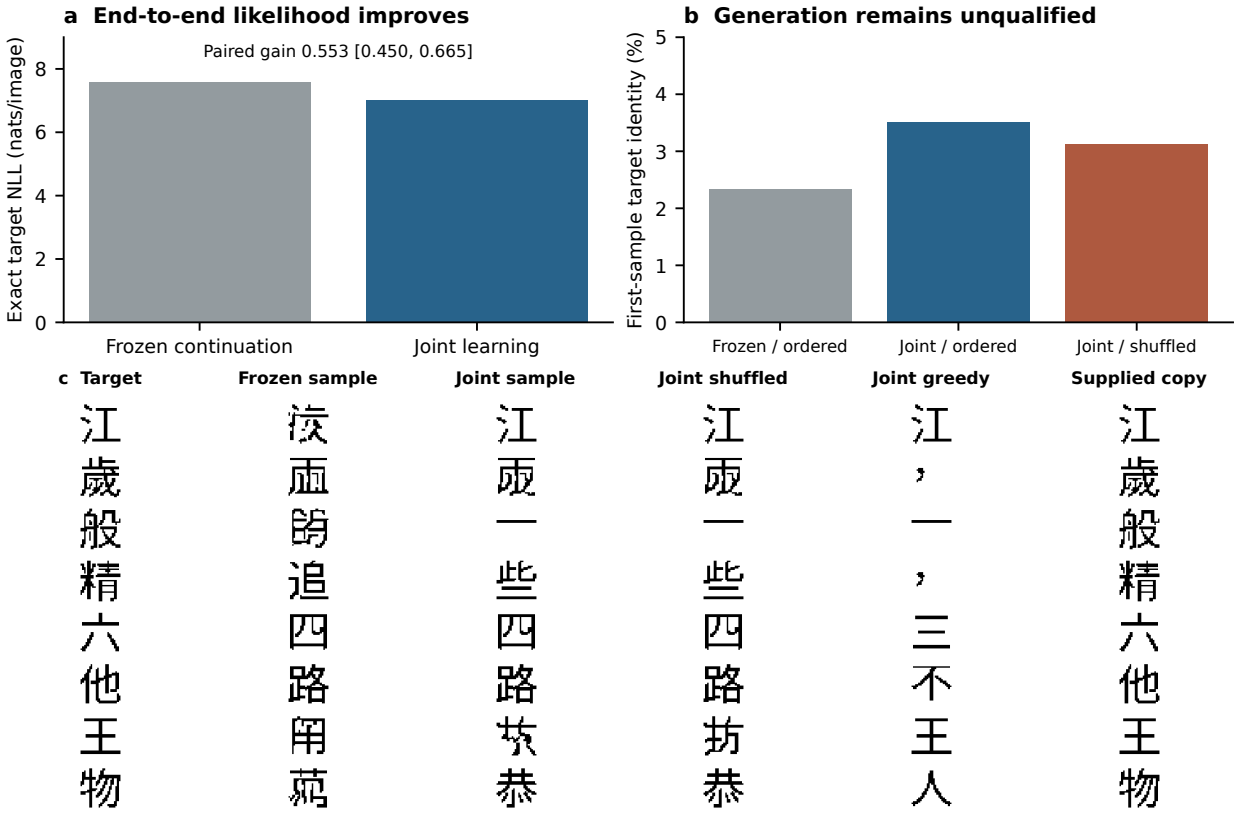


Fig. S2 | Matched end-to-end learning of exact raster probability. a, Joint reader–density training improves target NLL over the matched frozen-reader continuation; annotation gives the paired source-record interval. **b**, First-sample identity on 256 prefixes remains low under ordered and earlier-shuffled conditions. **c**, First eight fixed audit rows: target, frozen sample, joint sample, joint shuffled sample, joint greedy output and answer-supplied copying. All outputs are raw model rasters, without selection, bank replacement or OCR repair. The image panels expose errors that the likelihood improvement alone cannot resolve.

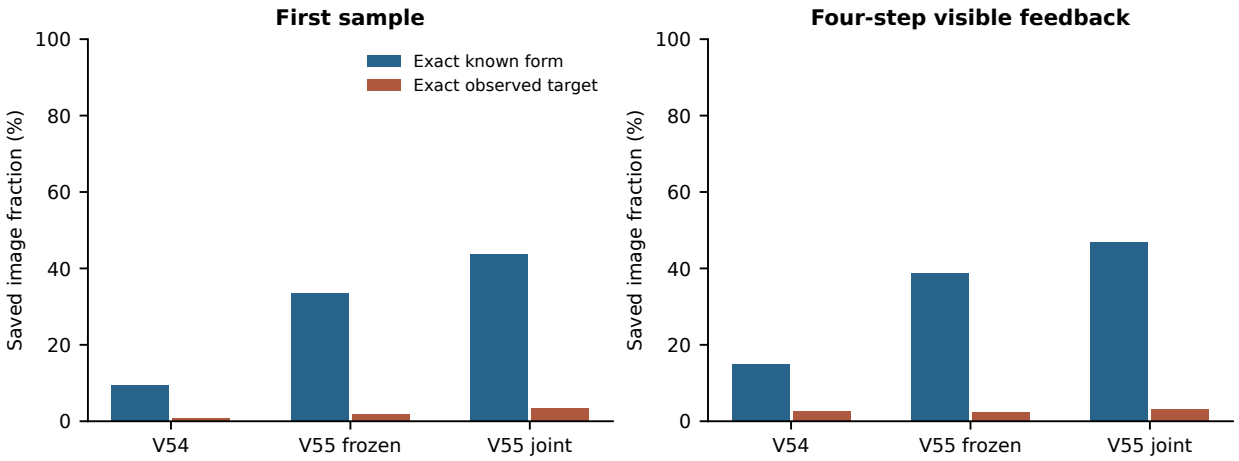


Fig. S3 | Exact canonical form does not imply correct continuation. Saved first samples and four-step feedback images are compared with a fixed canonical raster bank. Blue bars require exact pixel equality with any known form; brown bars require equality with the observed future image. All three checkpoints use the same bank. The paired uncertainty is stated in Note 14; the bars are descriptive fractions, not independent replications.

Supplementary Note 15: raw-pixel beam decoding

The V56 decoder changes search alone. V55 joint weights and the complete binary-image law remain fixed. A beam of eight prefixes expands the next binary pixel to ink or background, adds its exact conditional log probability and retains the eight best scores. The row model observes preceding rows; the masked within-row head observes preceding pixels. Parent gathering keeps row history and row conditions aligned. Stable ties prefer ink and then the earlier parent. Width one reproduces the original greedy rule. No persistent glyph inventory, target identity, image replacement, rejection filter or external model enters the decoder. Beam eight is approximate mode search, not exact global maximization over all 2^{1024} images.

A two-image numerical preflight found that the original high matmul precision gave incremental/parallel score differences above the frozen 10^{-3}-nat tolerance. An amendment recorded before production sets both beam and greedy controls to highest FP32 matmul precision. It changes no weights, windows, beam width, tolerance or quality gate. Production maximum score discrepancy is 2.10×10^{-5} nat and the width-one equality check passes.

The fixed audit uses 2,048 development windows, seed 20265620, 64 visible cells and eight future targets. Exact next-image matches are 238/2,048 for ordered beam, 202/2,048 for earlier-shuffled beam and 169/2,048 for ordered greedy: 11.621, 9.863 and 8.252%, respectively. Shuffling preserves the final four visible cells. The paired beam-minus-greedy gain is 3.369 points [2.351, 4.412]; ordered-minus-shuffled is 1.758 points [0.966, 2.552]. Intervals use 2,000 source-record bootstrap draws, seed 20265719, with 188 source records. All ordered and shuffled beam outputs exactly match a known canonical form. The post-generation bank has 9,934 distinct rasters and covers every target; it is descriptive, not the search space. Beam emits 127 distinct first forms, versus 116 greedy. Its mean self NLL is 3.108 versus 3.508 nats/image; this scores the selected output, not the true target, and is not a training gain.

Visible feedback uses the first 128 prefixes for eight steps. Exact observed image matches are 68/1,024 ordered beam, 51/1,024 shuffled beam and 50/1,024 greedy. The beam-minus-greedy difference is 1.758 points [0, 3.770], with 89 source records. Adjacent image repetition is 228/896 transitions (25.446%) for beam, versus one transition (0.112%) in the observed streams. Shuffled beam repeats 32.366% and greedy 17.746%. Mean distinct images per eight-step stream are 4.711 beam and 7.789 observed. Thus search improves immediate conditional choice while longer output remains repetitive (Fig. S4). Five of six gates pass; feedback repetition fails and the decision is `reject_decoder`.

All 256 supplied motor images still copy exactly. The audit takes 269.3 s with 0.154 GiB peak allocated CUDA memory on GPU 0, excluding previous training and data preparation. No new training or sealed evaluation is performed. This bounded search result does not establish question answering, instruction following, historical-script generation or capability-matched efficiency against a text language model.

Supplementary Note 16: grounded image-answer learning

The V57 model retains the V55 joint reader and exact raster writer, extending the allowed causal context from 64 to 512 cells. Parameter-free rotary positions require no weight-shape change; this extension does not itself establish trained long-context competence. Two 384×384 projections produce source-position attention from the final causal query and observed prompt states. A linear copy gate and a linear stop head give 295,682 new parameters, for 19,007,523 runtime parameters. The contrastive temperature remains unused and fixed. Both reader and density continue learning.

For observed source images X_i, attention a_{ti} and copy gate λ_t, the continued image distribution is

$$q_t(Y) = \lambda_t \sum_i a_{ti} \mathbf{1}[Y = X_i] + (1 - \lambda_t) r_t(Y). \tag{4}$$

Here r_t is the normalized exact raster density. Duplicate source images contribute summed mass. Equality uses lossless bit packing of observed binary pixels; there is no persistent symbolic identity or writing bank. A query cannot attend to source positions later than itself, including when natural training sequences are packed together. With stop hazard s_t, $p_t(\text{stop}) = s_t$ and $p_t(Y, \text{continue}) = (1 - s_t) q_t(Y)$. Training includes the terminal stop event after a reference answer. Reference length determines supervision masks, not a deployed input. Runtime receipts retain an inherited Bernoulli-density description; it describes r_t alone, not the complete mixture/event law.

CMRC 2018 supplies natural questions over Wikipedia paragraphs [12]. Pinned source revision and original annotation offsets are checked during data preparation. Whitespace is removed consistently, unsupported font forms are excluded, full rendered prompts are limited to 480 cells and retained answers to 16. No answer-centred crop is used. Answers must match the source substring and not occur verbatim in the question. The result contains 3,211 training questions over 1,134 paragraphs and 1,321 development questions over 421 paragraphs. Article-title and context-hash overlap are zero across splits. The data retain CC-BY-SA-4.0 attribution. Canonical rendering and thresholding match V55. No symbolic label is passed to a student method.

Both arms initialize from the same V55 EMA, train for 4,000 updates with batch four and use identical data RNGs. Three of four updates draw uniform training questions with replacement; the fourth retains 16 causal positions per natural stream, including 25% supplied-image motor conditions. Motor conditions use zero hidden states and exact observed fields, bypass copying and omit the stop loss. The no-copy arm fixes $\lambda = 0$; its source-attention heads are

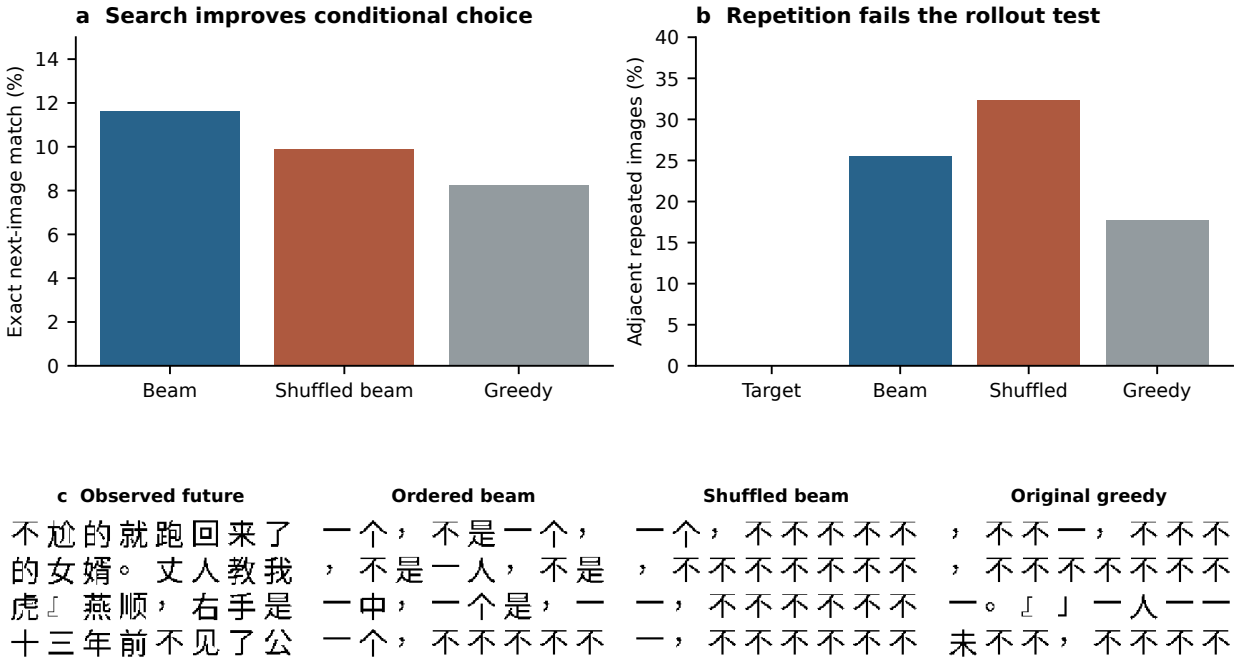


Fig. S4 | Pixel search improves next-image choice but fails rollout. a, Exact target-image matches on 2,048 fixed prefixes, with unchanged weights across ordered beam, earlier-shuffled beam and greedy controls. **b**, Adjacent raster repetition over eight-step feedback, compared with observed future streams. **c**, First four fixed audit streams: observed future and raw generated images. No examples are selected or repaired. Paired intervals are reported in Note 15; the feedback failure remains the decoder's qualification decision.

frozen. The first retained answer supervises event likelihood, while evaluation accepts every retained reference. Loss is mean exact event NLL over valid image and stop events, without the earlier image-only $1/1024$ rescaling.

AdamW rates are 3×10^{-5} for the reader, 10^{-4} for the density and 3×10^{-4} for new heads; betas are (0.9,0.95), weight decay 0.01, warmup 200 updates, cosine decay to 10%, norm clip one and EMA 0.999. BF16 training uses deterministic CUDA algorithms, math SDPA and fixed cuBLAS workspace configuration; reader dropout stays disabled. A four-update smoke matches interrupted two-plus-two training exactly for all model/EMA tensors and data, training, CPU and CUDA RNG states. A masked-copy backward regression test covers matches occurring only at forbidden future positions. Copy and no-copy training take 278.572 and 269.126 s, with 1.053 and 1.049 GiB peak allocated CUDA memory on GPU 0. These costs exclude previous foundation training and preparation.

Both final EMA checkpoints complete before development opens. Primary IDs are 256 samples without replacement from sorted development records, seed 20265720. The decoder stops at hazard ≥ 0.5; otherwise it selects the copy component at gate ≥ 0.5, sums attention over equal observed rasters and emits the largest-mass image. The free component uses pixel beam eight. This is hierarchical greedy decoding, not global mixture-mode search. The 17-cell safety cap is recorded separately from stopping. No answer location, reference length or candidate answer list is available at inference.

Exact answer match requires every output image, its order and learned stop to match a retained reference. Raster F1 measures multiset overlap of exact rasters, maximizing over references. It can credit reordered images and is not the official word-segmented CMRC metric. Primary copy exact match is 10/256 (3.906%), versus 1/256 with question pixels blanked and 0/256 no-copy. Copy F1 is 15.718%, versus 5.553% and 6.582%. Its paired gain over blanking is 10.165 points [7.043, 13.371]; over no-copy it is 9.136 [5.879, 12.483]. Intervals use 2,000 source-article draws, seed 20265819, retaining related questions together (198 primary articles). The blanking intervention removes question ink while retaining passage and framing; it is not an in-distribution alternative natural question.

All 393 shared-paragraph/disjoint-answer question pairs are also tested. Only one has both answers exactly correct (0.254%). Mean primary answer length is 10.457 cells, and 58/256 cases reach the safety cap. Every primary copy-arm image uses copying, with no free-branch emission. Teacher-forced mean event NLL across all 1,321 development questions is 1.329 copy versus 12.025 no-copy; mean sequence NLL is 8.377 versus 91.477. All 256 V55 supplied motor images remain exact. The full audit takes 1,117.1 s and peaks at 0.140 GiB allocated CUDA memory, including the slower 2,106-image free-writing control. Four of seven gates pass: boundary, clean split/resources,

motor and question-dependent F1. Exact answers, absolute F1 and paired-question correctness fail. The decision is `reject_reading_model`.

A post-result, zero-model-call diagnostic on the saved primary outputs finds 43/256 correct first images. Only 10/43 of those become exact complete answers; 14/256 outputs contain a complete reference prefix followed by unwanted additional images. Mean aligned correct prefix length is 0.750 cells; adjacent repetition is 2.230%. These descriptive observations do not alter the locked decision. They motivate testing answer-level structure alongside better evidence selection. All results remain limited to canonical, answerable, short-paragraph reading; independent seed replication, raw page segmentation, historical writing and general language generation are unestablished.

Supplementary Note 17: complete observed image blocks

V58 continues the completed V57 copy EMA with a normalized distribution over all contiguous observed prompt blocks $X_{i:j}$, where $1 \le j-i+1 \le 16$. For causal states h_i and final prompt state h_L, define

$$s_{ij} = \frac{(W_q^s h_L)^\top W_k^s h_i + (W_q^e h_L)^\top W_k^e h_j}{\sqrt{384}} + b_{j-i+1}, \qquad \pi_{ij} = \frac{e^{s_{ij}}}{\sum_{uv} e^{s_{uv}}}.$$

The answer-image law is

$$p(Y \mid X) = \sum_{ij:\, X_{i:j}=Y} \pi_{ij}.$$

Equality is lossless raster equality, computed from binary pixels; indices address only the current prompt. There is no persistent glyph inventory. Training minimizes exact negative log block mass, divided by reference cell count plus one boundary event across the batch. Annotated source offsets do not enter the loss. Mode decoding sums mass over equal sequences and chooses the most probable sequence, with first-source tie breaking. No reference length, answer list, OCR or output repair enters generation. The predicted end boundary terminates a nonempty block; the 16-cell support is fixed before training. The retained copy/free/stop continuation path remains separate.

Four new 384-by-384 projections and sixteen length biases bring the runtime to 19,597,363 parameters. Projection pairs initialize from V57 copy attention; the matched glyph control constructs and freezes the same new heads. Both start from identical V57 copy EMA weights and make 4,000 updates, batch four, using the previous three-to-one QA/natural-motor schedule, optimizer rates and deterministic settings. Seeds are 20265810 (training) and 20265811 (data). A four-update uninterrupted versus two-plus-two resumed GPU run matches all model/EMA tensors and data, training, CPU and CUDA RNG states exactly. Block training takes 210.995 s, 0.967 GiB peak allocated memory; the glyph control takes 302.237 s, 1.056 GiB. Earlier foundation costs are excluded.

Both final checkpoints precede development inspection. Primary IDs are the same 256 as V57, explicitly reused development, not a sealed final test. Exact answers are 25/256 (9.766%) block, 3 blank-question and 21 matched glyph control. F1 is 27.033%, 9.169% and 22.140%. Article-grouped bootstrap uses 2,000 draws, seed 20265919. Block-minus-control F1 is 4.893 points [0.469, 9.047]; exact match is 1.563 [-1.509, 4.652]. Normal-minus-blank F1 is 17.864 [13.438, 22.289]. Only 4/393 pairs have both answers correct; the paired gate fails and the registered decision remains `reject_reading_model`. All 256 supplied motor images remain exact. Mean conditional block sequence NLL is 6.505, versus 9.552 for glyph-event sequence likelihood; these laws have different supports and must not be treated as interchangeable language perplexities. The audit takes 98.951 s at 0.142 GiB allocated memory. The 78,772,107-byte inference-only export reproduces native prompt-PNG answers and events exactly. Raw first audit failures appear in Figure 5.

Supplementary Note 18: frozen visual foundation probe

V59 pins MIXAR-477M stage-2 revision `c5d5bcfbad2f290296fd83810558f396c6a6333a`, reusing its shared Apache-2.0 checkpoint. Author-reported multilingual pretraining and its four-H200 compute are external work, not part of single-4090 training here. Both it and the frozen V58 reader consume identical Noto CJK 26-pixel glyphs inside binary 32-pixel cells. This differs from MIXAR's native rendering. An odd-length prompt gets a blank final cell for two-cell pairing; readout candidates exclude that extra cell. Both encoders supply states after each complete pair, repeated over its two source cells.

A fixed CPU-QR semi-orthogonal projection maps MIXAR's 896 states to 384 coordinates, seed 20265900; student states already have width 384. Both append 128 projected normalized-DCT coordinates, seed 20265901, plus two within-pair phase channels. Semantic and fine-image parts are separately RMS-normalized. The 514-dimensional features are quantized to FP16 for caching and readout. All train, development and blank-question features are prepared without reading answer annotations. Recomputed quantized features agree exactly.

Each fresh head has four 514-to-384 projections and sixteen biases, totaling 789,520 learned parameters. Both train 4,000 updates, batch eight, with seeds 20265910/20265911, AdamW rate 3×10^{-4}, betas (0.9,0.95), decay 0.05, norm clip one, warmup 200, cosine to 10%, and EMA 0.999. Frozen encoders and native free writers are unchanged. An interrupted/resumed GPU preflight matches weights, EMA and RNG states exactly. Features take 184.781 s and 1.906

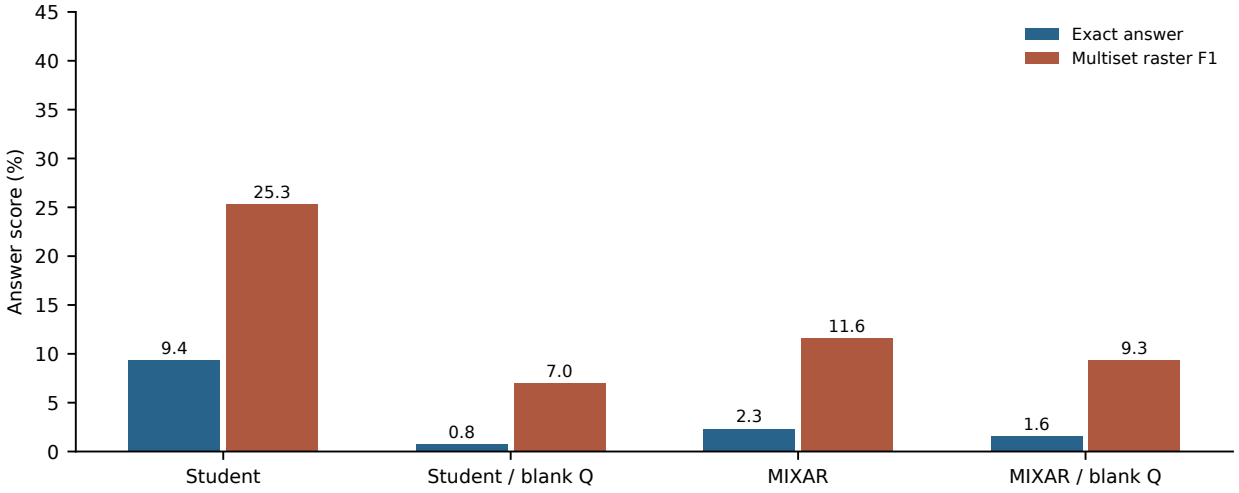


Fig. S5 | Frozen pretrained features do not improve this reading interface. Identical learned block heads consume frozen student or MIXAR features, with normal and blank-question prompts. All scores use the fixed 256 questions. External pretraining and rendering shift prevent a causal attribution to capacity or pretraining alone.

GiB; student and MIXAR readouts take 106.593 and 109.257 s, each at 0.170 GiB. These resource numbers exclude external pretraining.

Both completed heads precede inspection of the same reused 256 development IDs and 393 pairs. Student exact/F1 scores are 9.375/25.340%; MIXAR scores 2.344/11.588%. MIXAR-minus-student F1 is -13.752 points $[-18.247,-9.094]$, using 2,000 article draws, seed 20266019. Normal-minus-blank F1 is 2.246 [0.415,4.378] for MIXAR, versus 18.343 [13.949,22.519] for the student. Both questions are exact in zero versus twelve pairs. Three of seven gates pass; the frozen-foundation probe fails. The 117.085-s, 0.094-GiB audit uses cached readouts and excludes online encoder inference. Figure S5 reports all four conditions. Capacity, pretraining, prior task training and rendering differ; this is a scoped foundation probe, not a general ranking of encoders or native-font MIXAR competence.

Supplementary Note 19: question-first causal reading

V60 tests whether passage states should be conditioned on the question before evidence selection. The context-first host layout places passage X before question Q; the intervention reverses these two blocks, yielding causal passage states $h_i = f(Q, X_{\leq i})$. Framing and total visible length are unchanged. Both models start from identical V58 block EMA weights and retain the 19,597,363-parameter architecture, block law, image-only supervision, mode decoder and continuation writer. All visible prompt positions remain candidates. There is no answer offset or answer-aware cropping.

Each arm makes 4,000 updates with the same V58 batch-four schedule, optimizer rates, warmup, cosine, clipping, EMA and deterministic BF16/math-SDPA regime. Training/data seeds are 20266010/20266011. The new prompt-order helper is tested for exact preservation of source characters and correct blanking positions. An uninterrupted versus two-plus-two resumed GPU preflight matches all model/EMA tensors and RNG states exactly. Both production arms finish before their development scores open. Context-first and question-first training take 214.699 and 225.207 s, each at 0.966 GiB peak allocated CUDA memory. Cost remains incremental to the same previously trained foundation.

The same 256 primary IDs and 393 paragraph pairs are explicitly reused. Question pixels are blanked at each arm's actual question positions, retaining the paragraph, length and framing. Exact answers are 27 context-first and 21 question-first, versus three and two for the corresponding blank controls. F1 is 26.908% and 23.765%, versus 7.572% and 9.098% blanked. With 2,000 article-grouped bootstrap draws, seed 20266119, question-first minus context-first F1 is -3.143 points $[-6.544,0.446]$; exact match is -2.344 $[-5.490,0.810]$. Neither difference resolves zero. The question-first normal-minus-blank F1 gain is 14.667 [10.873,18.809]. Both answers are exact in 5/393 question-first pairs, versus 16/393 context-first pairs.

All 256 supplied motor images remain exact for both arms. The complete audit takes 192.723 s at 0.184 GiB allocated CUDA memory. Four of eight registered gates pass: boundary, clean split/resources, question-dependent F1 and motor. The gain, absolute accuracy, absolute F1 and paired-question gates fail, so the decision is `reject_question_first_intervention`. This is one matched training seed and a task-trained, context-first foundation; neither a universal ordering result nor a final generalization test is established.

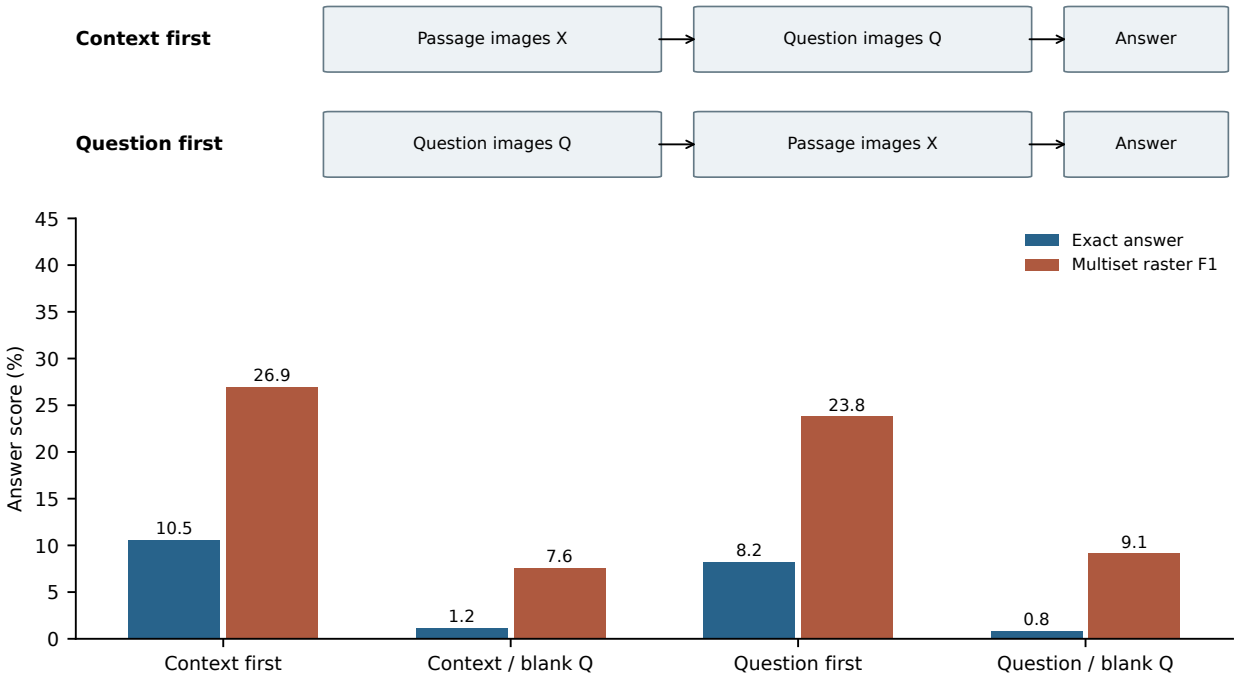


Fig. S6 | Changing input order is insufficient. Matched models receive the same source writing with the question before or after the passage. Question blanking uses each layout's actual positions. The question-first F1 difference is unresolved and its registered reading criteria fail.

Supplementary Note 20: full native pixel-language adaptation

The complete released MIXAR-477M[9] image-language model is adapted using the pinned release identified by its full revision, weight SHA-256 and architecture-code hashes in the accompanying evidence. This is supervised adaptation of external pretraining, not local foundation training from scratch. All 477,145,984 contextual and image-output parameters train. The source data are the same 3,211 training and 1,321 development questions used for the preceding reading experiments, with no article or context-hash overlap. Development is explicitly reused. Complete prompts contain at most 480 cells and first-reference training answers at most 16 cells.

Noto CJK glyphs at 26 pixels occupy 32-pixel binary cells, a rendering shift from the author's native bundled-font interface. Odd prompt/answer lengths receive one blank cell for alignment to 32-by-64 native blocks. Two all-white blocks follow each target answer. Native input uses white $+1$ and ink -1; the target Bernoulli probability describes white pixels. Spatial prompt bounds are known, but strings, codepoint IDs and source-answer offsets never enter the actor. The causal answer-only objective is

$$\mathscr{L} = -\frac{1}{|A|P}\sum_{t\in A}\sum_{i=1}^{P}\left[y_{ti}\log\sigma(a_{t-1,i}) + (1-y_{ti})\log(1-\sigma(a_{t-1,i}))\right], \tag{5}$$

where $P = 2048$, A contains answer and termination blocks, and state $t-1$ depends only on preceding physical images. Prompt labels contribute no loss.

The registered run uses 1,000 updates, microbatch one accumulated eight times, uniform replacement sampling, AdamW 2×10^{-5}, betas $(0.9, 0.95)$, weight decay 0.01, warmup 50, cosine decay to 10%, norm clip one and EMA 0.99. Seeds are 20266110 for training and 20266111 for data. FP32 weights, optimizer and EMA coexist with BF16 forward/backward, deterministic algorithms, math attention and nonreentrant activation checkpointing. The final EMA is used without development checkpoint selection. A real four-update interrupted preflight reproduces model, EMA, optimizer and all RNG states exactly, with finite gradients throughout. Altering future answer pixels changes earlier logits by exactly zero in the recorded causality check.

Training through the final update takes 1,501.138 s, with 9.912 GiB peak allocated memory; loading takes 2.525 s. Cache preparation, final serialization and external pretraining are separate. All work uses physical GPU 0. Greedy decoding thresholds each white logit at zero, feeds back the generated image and stops at the first exactly white block, otherwise capping at ten new blocks. Only terminal blocks and entirely blank trailing cells are removed. No malformed image is repaired or snapped to a character.

The locked audit compares adapted and released-base actors with normal and blank-question inputs on the same 256 primary IDs, and tests all 393 shared-paragraph question pairs for the adapted actor. All 1,321 development examples

contribute likelihood. Pixel NLL decreases from 0.684156 to 0.154814 nats, yet exact answers are zero and exact-raster multiset F1 is 0.149%. Foreground F1 is 12.264%, compared with 8.473% blanked and zero for the released base under this shifted interface. Article-grouped bootstrap with 2,000 draws, seed 20266219, gives the adapted-minus-blank foreground difference as 3.791 points [2.521,5.056]. The exact-raster F1 difference is 0.149 [0,0.372], unresolved. Both-correct pair accuracy is 0/393.

No primary output is empty or capped for the adapted actor; the released base returns an empty answer on every normal primary prompt. The adapted mean length is 3.047 cells. Five of ten gates pass: image-only runtime, clean split/finite resources, full backward/resume/causality, independent replay, and capped-output rate. Exact answers, raster F1, adaptation gain, question-dependent raster F1 and paired correctness fail. Audit time is 145.690 s excluding loading, at 1.832 GiB allocated memory. The registered decision is `reject_native_free_reading`.

A learned-weight deployment bundle includes configuration and local custom architecture code. A fresh offline process with empty model and code caches reproduces actual answer tensors, PNG bytes and stopping events. It loads no original release weights and calls no external model, tokenizer or OCR. Exact replay is an implementation result, separate from language quality.

An auxiliary transcription plan was registered before development outputs were inspected. CPU Tesseract 5.3.4 receives anonymous answer-only images, enlarged fourfold with a white margin, without questions, references or model names. It recognizes only 40/256 retained reference images exactly, a 15.625% calibration rate. No generated answer is transcribed exactly, but the poor reference calibration precludes treating this as a reliable semantic score. The primary raw-image failure and strict gates remain unchanged. One seed, the shifted font interface, limited training data and absent general-language evaluation constrain the interpretation.

Supplementary Note 21: conditional whole-image density

The matched experiment tests a latent expert shared across a complete native image block. Let h_t be the visual transformer's causal state after the visible prefix and let $y \in \{0,1\}^{2048}$ denote the next 32×64 bitmap, with one denoting white in this native interface. The normalized law is

$$p_\theta(y \mid h_t) = \sum_{k=1}^{K} \pi_{\theta k}(h_t) \prod_{i=1}^{2048} q_{\theta ki}(h_t)^{y_i} [1 - q_{\theta ki}(h_t)]^{1-y_i}, \qquad q_{\theta ki} = \sigma(a_{\theta ki}), \tag{6}$$

where the gate is a softmax. Each expert is a neural function of the current visual state, without a fixed character identity or stored target bitmap. Marginalizing the expert retains a proper probability law. For distinct pixels, conditional covariance is

$$\mathrm{Cov}(Y_i, Y_j \mid h_t) = \sum_k \pi_k q_{ki} q_{kj} - \Big(\sum_k \pi_k q_{ki}\Big)\Big(\sum_k \pi_k q_{kj}\Big). \tag{7}$$

It can be nonzero even though pixels are independent within one expert. This elementary property permits coherent alternatives; it does not prove that training discovers valid writing or useful answers.

The objective is exact negative log joint likelihood, computed with `logsumexp`, divided by 2,048 physical pixels. It averages only answer and two blank termination blocks within each example, then averages over accumulated examples. Teacher forcing uses the same one-block causal shift as Note 20. Prompt pixels enter the context without a prompt loss. There is no transcript, answer-span offset or Unicode target inside the actor.

The single-head control has 477,145,984 parameters; eight experts have 489,998,208. Both initialize from the same externally pretrained MIXAR revision and use the unchanged Noto-26 cell interface, 3,211 training questions and 1,321 article-disjoint development questions. The eight-head projection repeats the released head with zero-mean perturbations of scale 0.01, seed 20266210, and a zero-initialized gate. Every backbone and head parameter receives gradients. The comparison matches examples, optimization and update budget, rather than parameter count or output computation.

Each arm uses 1,000 updates, microbatch one, accumulation eight, AdamW at 2×10^{-5} with betas $(0.9, 0.95)$, decay 0.01, 50-update warmup, cosine decay to 10%, norm clip one and EMA 0.99. Weights, optimizer and EMA are FP32; forward/backward use BF16 with deterministic algorithms, eager/math attention and nonreentrant activation checkpointing. Training and data seeds are 20266220 and 20266221. Complete sampled-example orders must match exactly. Only the final EMA is audited. Both arms run sequentially on physical GPU 0, under a 20 GiB allocated-memory and four-hour ceiling per phase.

Generation thresholds each expert into one complete candidate bitmap, scores all candidates under the full mixture and emits the highest-scoring candidate. Searching component modes is approximate MAP; it is not a guarantee of the global mixture mode. Actual emitted pixels become subsequent context. The first exactly white native block ends generation, otherwise ten blocks cap the continuation. Only the detected terminal block and wholly blank trailing cells are removed. Malformed nonblank strokes are preserved. The deployed bundle has no tokenizer, OCR, glyph inventory or external language-model call.

The primary audit retains the fixed 256 questions, blank-question controls, all 393 shared-paragraph pairs and all-development likelihood. Paired intervals resample articles 2,000 times, seed 20266229. An intervention pass requires positive lower intervals for both K8-minus-K1 strict-raster F1 and K8 normal-minus-blank F1, fewer than 20% capped outputs and all implementation checks. Each arm's separate bounded reading gate requires at least 10% exact answers, 25% raster F1, 2% both-correct pairs, a positive question effect, fewer than 20% caps and implementation checks. Expert-use counts and gate entropy are descriptive; they cannot select a new checkpoint or change these gates. Reused development, one seed, the shifted font interface and external pretraining limit the conclusions.

Both arms complete their registered budget and all replay checks. K=1 takes 1,887.5 s at 9.904 GiB peak allocated memory; K=8 takes 1,972.8 s at 10.174 GiB. Their complete training-example orders match exactly. The locked audit takes 527.4 s excluding model loading, at 1.904 GiB allocated memory. Four-update uninterrupted versus two-plus-resumed-two preflights match model, EMA, optimizer and every RNG state exactly for each arm. Future-pixel changes have zero effect on earlier predictions. Strict-FP32 cache errors remain below 2.1×10^{-5}; larger TF32 discrepancies are retained separately rather than hidden by changing the production recipe.

All-development NLL is 0.1542 for K=1 and 0.1248 for K=8. Primary strict-raster F1 is 0.234% versus 0.815%, a 0.580-point gain [0.089, 1.094]. Foreground F1 is 12.205% versus 15.827%, a 3.623-point gain [1.815, 5.554]. Both arms have zero exact primary answers and zero both-correct pairs. K=8's normal-minus-blank raster difference is 0.295 points [−0.006, 0.623]; its foreground difference is −0.044 points [−0.629, 0.509]. Neither resolves question dependence. Seven of eight intervention gates pass; both arms pass six of ten bounded reading gates. The registered decision is `reject_density_intervention`; neither arm qualifies as free reading.

K=8 selects experts 0, 1, 2, 4 and 7 on respectively 223, 4, 106, 605 and 3 primary generation steps, including termination. Mean gate entropy is 1.812 nats, compared with $\log 8 = 2.079$ for a uniform gate. The primary outputs all terminate before the cap, while one blank-question output reaches the cap. These descriptive observations do not identify semantic expert roles or justify selecting different parameters. The result supports improved bitmap fit at this budget, while the required question-dependent language remains unestablished.

Auxiliary reader and post-hoc transcript sensitivity

A separately preregistered, answer-only Qwen3-VL 8B reader transcribes 122/128 training-reference images exactly (95.313%), returns every one of 16 blank and 16 random-ink controls empty, and produces valid JSON throughout calibration. The fixed instruction and anonymous images disclose neither questions nor reference answers to the reader. Three responses among 1,015 unique actor images are invalid, including two that reach the registered 256-output-token limit. The original diagnostic therefore reports `invalid_actor_transcription`; it has no complete fixed-case score.

An explicitly post-hoc sensitivity analysis reuses the first responses without retry, repair or new model calls. It retains all 256 cases in each condition and assigns each invalid bounded score possible range $[0,1]$. The resulting full-case means are below; ranges are identification intervals, not confidence intervals. Exact text means normalized transcript agreement with any retained reference. Character score is one minus normalized edit error, clipped to $[0,1]$.

Condition	Valid / unknown	Exact text (%)	Character score (%)
K=1	254 / 2	0.781–1.563	1.388–2.169
K=1, blank Q	255 / 1	0.000–0.391	0.186–0.577
K=8	256 / 0	0.000	2.806
K=8, blank Q	256 / 0	0.000	2.945

K=8's normal-minus-blank character difference is −0.139 percentage points, with article-bootstrap sensitivity envelope [−0.962, 0.638]. The envelope uses 2,000 article resamples, seed 20266379, taking the lower endpoint's 2.5th percentile and upper endpoint's 97.5th percentile. It covers the unknown-score assignments within this resampling scheme, not systematic reader errors, training-seed variation or reused-development selection. It establishes no new model qualification and changes neither failed decision. The external reader makes 1,172 calls over 1,092.1 s of owned runtime including cleanup; maximum resident GPU memory sampled after requests is 7,278 MiB. These costs are separate from actor inference and training. The evaluator server and runner exit, its port closes and GPU 0 memory is released.

Supplementary Note 22: matched question–answer assignment

For two observed question images x_0, x_1 about the same paragraph and two accepted answer-image streams y_0, y_1, define e_{ij} as mean answer and termination-pixel NLL for y_j conditioned on x_i. Each actor scores all four streams using

the complete image generator. Let

$$D = e_{01} + e_{10} - e_{00} - e_{11}, \tag{8}$$

$$\mathscr{L}_{\text{positive}} = (e_{00} + e_{11})/2, \tag{9}$$

$$\mathscr{L}_{\text{assign}} = \tau \log(1 + \exp[-D/\tau]), \qquad \tau = 0.01, \tag{10}$$

$$\mathscr{L} = \mathscr{L}_{\text{positive}} + \lambda \mathscr{L}_{\text{assign}}. \tag{11}$$

Aligned uses $\lambda = 1$ and control $\lambda = 0$. Additive question-only and answer-only energy offsets cancel from D. Different answer lengths receive their own normalization, so D is not a joint log-odds or mutual-information estimate. The algebra does not identify linguistic meaning.

There are 3,793 training-only pairs from 939 paragraphs and 3,016 questions. Accepted answer text sets and canonical reference images are distinct within each pair; two pairs with shared references are excluded. Sampling chooses paragraphs uniformly, then a pair within the paragraph. Both actors start from the same learned V62 eight-expert EMA, retain 489,998,208 parameters and use the unchanged Noto-26 interface and approximate component-mode decoder. No reference bank, OCR or teacher is deployed. All 2,000 update journals, containing four pair draws each, match byte for byte.

Microbatch is one pair with accumulation four. Each pair forms two positive and two swapped likelihood streams. AdamW uses learning rate 2×10^{-5}, betas $(0.9, 0.95)$, decay 0.01, 100-update warmup, cosine decay to 10%, clip one and EMA 0.99. Model/optimizer/EMA weights are FP32; forward/backward use BF16, deterministic eager/math attention and nonreentrant checkpointing. Model and data seeds are 20266320 and 20266321. Only final EMA is audited. Both arms execute sequentially on physical GPU 0 under eight-hour and 20-GiB allocated-memory ceilings per phase.

Supplementary Table 4: completed question–answer assignment. Image F1 values are percentages.

Arm	Exact / 256	Raster F1	Foreground F1	Both / 393
Control	2	3.229	20.227	0
Aligned	1	2.417	17.219	0

Control and aligned all-development NLLs are 0.113901 and 0.114186 nats per pixel. Aligned minus control raster F1 is -0.812 points $[-1.504, -0.202]$; foreground F1 differs by -3.008 points $[-3.978, -2.014]$. Normal-minus-blank raster effects are 1.765 points [0.504,3.251] for control and 1.677 [0.492,3.033] for aligned. All intervals resample source articles 2,000 times, seed 20266329, over the same 256 primary cases from 198 articles.

Each actor passes seven of ten reading gates. Both fail the unchanged 10% exact-answer, 25% raster-F1 and 2% both-correct-pair requirements. Question dependence, capped outputs and implementation checks pass. The intervention requires an aligned reading pass and a positive lower interval for aligned-minus-control raster F1; both requirements fail. The decision is `reject_assignment_intervention`. No decoder, checkpoint selection, threshold or primary case changes after scoring.

The separate accepted-reference diagnostic assigns the pair correctly in 324/393 cases for control and 330/393 for aligned. Both individual energy ranks are correct in 82/393 and 109/393. These are descriptive comparisons without registered difference intervals, and do not evaluate free writing. Training takes 7,355.156 and 6,737.336 s at 10.301 GiB peak allocated memory. The audit takes 582.104 s excluding loading, at 1.904 GiB. Timing excludes loading, final serialization and external foundation pretraining.

The preflight passes exact uninterrupted/resumed optimizer, EMA and RNG comparison, finite complete-backbone and assignment-only gradients, causal masking and physical stream assembly. Largest-pair allocation is 9.197 GiB. Same-shape future-pixel effects are zero; strict-FP32 cached-prefix error is 6.485×10^{-5} against a 2×10^{-4} bound. Both standalone bundles reproduce tensors, PNG bytes and stopping events with networking disabled, empty caches and no repository source path. Reused development, one seed, external pretraining and strict-raster limitations remain explicit. Sealed final partitions are unopened.

Supplementary Note 23: inherited continuous writing actor

A separately executed successor retains the existing authenticated Qwen3-VL vision/contextual archive[17], omits its vocabulary matrices and attaches a continuous physical-image writer (Fig. S7). It has 7,548,043,521 parameters: 7,522,463,984 inherited/frozen and 25,579,537 trainable. The trainable subset includes rank-eight query/value/output adapters in 36 contextual layers, two continuous writing queries, a 19,374,096-parameter six-layer flow writer and a binary stop classifier. All 27 vision layers and three DeepStack feature sets remain present. The actor accepts observed grayscale image tensors and completed 32×64 blocks; it requires no character segmentation or alphabet.

Teacher forcing places each query pair before its answer image. The observed prompt may be bidirectionally image-encoded, while each completed answer block is an independent encoder image. A causal contextual mask

prevents current and future answer pixels from entering a prediction. Cached generation rereads only completed generated blocks. Explicit spatial/temporal rotary coordinates preserve image layout; sequential mask coordinates carry no character identity.

The writer models all 2,048 pixels jointly. With Gaussian U, uniform t and signed target image Y, its flow objective is

$$Y_t = (1-t)U + tY, \qquad \mathscr{L}_{\text{flow}} = \mathbb{E}\|v_\theta(Y_t,t,h) - (Y-U)\|_2^2. \tag{12}$$

Explicit midpoint integration gives a continuous raster; zero thresholding produces binary output. A class-balanced binary stop objective is separate from white-space generation. Its sigmoid score has no independent calibration. No token IDs, OCR, font renderer, glyph bank or external model call enters actor inference. Encoded text remains permissible in offline dataset creation.

All 19 complete-model execution checks pass on GPU 0 at 17.181 GiB allocated and 18.053 GiB reserved peak memory. Geometry inspection covers 12,976 complete training examples from CMRC and Dolly instructions [12,18]. The largest contextual case has 918 positions, a 480×1024 prompt and 109 complete target blocks; the largest prompt is 512×1024. Both receive full contextual backward with Adam moments allocated, while 16 uniformly sampled target blocks receive writer loss. No prompt or answer history is truncated. Every trainable gradient is finite, and nonzero gradients reach first/last contextual adapters and the writing head. Restoring one saved optimizer update reproduces the next writer/adapters, optimizer, sampling and global RNG states exactly. Same-shape future-image and encoder-isolation interventions have exactly zero effect.

Two retained attempts fail cached/full-prefix comparison at relative maximum errors 0.18524 and 0.09815. The accepted implementation keeps contextual weight storage in BF16 but evaluates contextual projections, attention and residuals in FP32. Layer checkpointing avoids retaining a full FP32 body. Vision weights/computation are FP32; writer contractions may use BF16 autocast. The complete contextual call disables ambient autocast, verified at actual attention entry. Cached/full-prefix error becomes 3.910×10^{-4} absolute and 1.210×10^{-5} relative, below the unchanged 0.02 relative bound. These casts do not recover precision lost in the source quantization.

A separate offline helper temporarily reads source token rows and an output head to test the contextual initialization against the source engine. Across eight short text-only prompts, it matches 8/8 initial and 9/9 returned token choices, with mean KL 0.000264 nat after grouping reference top-ten tokens and the outside mass. This limited check is not full engine equivalence or image-language understanding. The helper vocabulary is never attached to the actor, and its owned reference service exits before native model loading. Its 103.832-s runtime and 18.701-GiB allocation include the temporary head and are separate from the image-actor preflight.

The preflight contains two optimizer updates and replay; its output is noise. It provides no writing or language quality score. A separate writer-only curriculum then trains 19,374,096 parameters for 10,000 updates on balanced Chinese/English blocks from 12,976 training examples. With the final EMA fixed, 128 development blocks yield foreground F1 of 34.000% under correct visual conditions and 21.420% with cyclically shuffled conditions, a gain of 12.581 percentage points. Neither route produces an exact block. Both registered component requirements, F1 at least 90% and gain at least 20 points, fail. Training peaks at 15.801 GiB; the resumed evaluation's separately reset peak is 15.514 GiB. The first evaluation attempt stops before output because its directory is missing. Creating that directory and resuming unchanged code at update 10,000 leaves final weights and training journals byte-identical, with no further learning. All 256 generated-image scores are recomputed from raw pixels. These are source-image reconstruction measurements on reused development data, not prompt-to-answer language. Independent autonomous answers, historical-form behavior and capability-matched efficiency remain required. External pretraining is inherited work, not training performed from scratch on the local GPU.

Reproduction

From the repository root, use the following entry points. All use only GPU 0.

```
CUDA_VISIBLE_DEVICES=0 PYTHONPATH=.  python
scripts/train_causal_visual_block_diffusion_v53.py -device cuda:0

CUDA_VISIBLE_DEVICES=0 PYTHONPATH=.  python
scripts/eval_causal_visual_block_diffusion_v53.py -device cuda:0
```

Completed output directories are protected against accidental replacement. For the exact density, with the same GPU selection, use

```
scripts/train_exact_raster_density_v54.py
scripts/eval_exact_raster_density_v54.py
```

Independent image inference and export use

```
scripts/infer_exact_raster_density_v54.py.
```

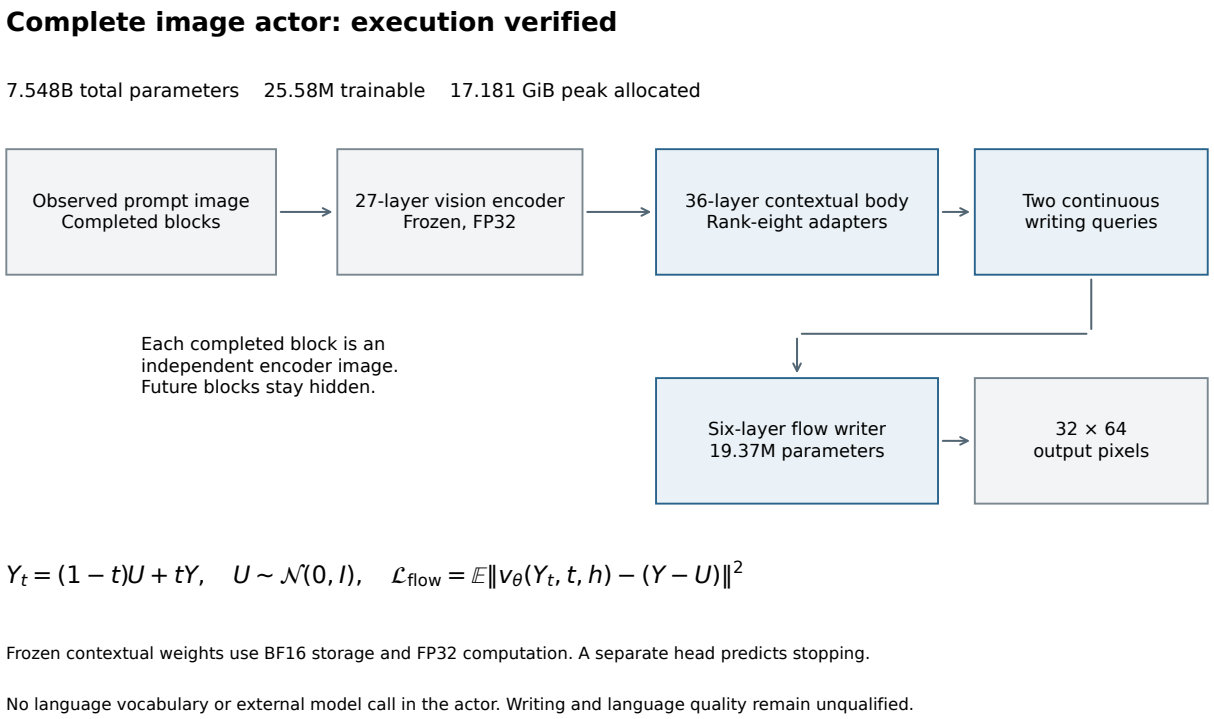


Fig. S7 | An inherited image actor executes within one GPU, with language quality unqualified. Observed prompt images and independently encoded completed blocks enter the retained visual/contextual body. Two continuous queries condition a flow over physical writing pixels. Blue modules contain trainable components; the contextual body also retains frozen inherited weights. A separate classifier predicts stopping. Parameter counts and memory describe actual complete-model execution. The diagram contains no generated-language example or quality claim.

For the matched continuation, run the following with `-arm frozen` and then `-arm joint`, completing both before the evaluator:

```
scripts/train_joint_exact_raster_v55.py
scripts/eval_joint_exact_raster_v55.py
```

The independent runtime uses `scripts/infer_joint_exact_raster_v55.py`. The form diagnostic and locked pixel decoder audit use

```
scripts/diagnose_raster_form_validity.py
scripts/eval_raster_beam_v56.py.
```

Independent pixel-beam inference uses `scripts/infer_raster_beam.py`. For grounded visual reading, complete both training arms before auditing:

```
scripts/train_visual_copy_language_v57.py -arm copy
scripts/train_visual_copy_language_v57.py -arm no_copy
scripts/eval_visual_copy_language_v57.py.
```

Native answers and self-contained export use

```
scripts/infer_visual_copy_language.py.
```

For complete image blocks, train both arms before the audit:

```
scripts/train_visual_answer_blocks_v58.py
scripts/eval_visual_answer_blocks_v58.py.
```

Native inference uses

```
scripts/infer_visual_answer_blocks.py.
```

Frozen-feature preparation, matched readout training and audit use the `cache_frozen_visual_features_v59`, `train_frozen_visual_readout_v59` and `eval_frozen_visual_readout_v59` scripts. The matched input-order experiment runs both orders before auditing:

```
scripts/train_question_first_reading_v60.py
-order context
-order question
scripts/eval_question_first_reading_v60.py.
```

Native pixel preparation, full adaptation and locked audit use the existing Python 3.12 MIXAR runtime, with `CUDA_VISIBLE_DEVICES=0`:

```
scripts/prepare_native_pixel_qa_v61.py
scripts/train_native_pixel_instruction_v61.py
scripts/verify_native_pixel_export_v61.py
scripts/eval_native_pixel_instruction_v61.py.
```

Independent image inference uses `scripts/infer_native_pixel_language.py`. The matched whole-image experiment uses the persistent sequential coordinator

```
scripts/run_correlated_native_v62.py.
```

It verifies both complete training reports, standalone offline replays and the locked audit in that order. Independent image inference uses `scripts/infer_correlated_native_language.py`; the exported bundle's `run.py` does not need the repository or the original foundation checkpoint. The optional, separately calibrated vision diagnostic is `scripts/audit_correlated_native_transcripts_v62.py`. That reader stays outside the actor and changes no primary gate. The complete assignment comparison uses `scripts/run_question_assignment_v63.py`; its independent raw-score verification is `scripts/verify_question_assignment_result_v63.py`. The separate successor preflight is `scripts/check_grounded_image_actor.py`, and its helper-only source check is `scripts/check_inherited_context.py`. The writing-only curriculum is specified in `references/grounded_writer_learning_protocol.md`; it is not an additional completed language result.

References

1. Tai, Y., Liao, X., Suglia, A. & Vergari, A. PIXAR: Auto-Regressive Language Modeling in Pixel Space. *arXiv* 2401.03321 (2024). https://arxiv.org/abs/2401.03321.
2. Li, T., Tian, Y., Li, H., Deng, M. & He, K. Autoregressive Image Generation without Vector Quantization. *Advances in Neural Information Processing Systems* (2024). https://arxiv.org/abs/2406.11838.
3. Chen, B. et al. Diffusion Forcing: Next-token Prediction Meets Full-Sequence Diffusion. *arXiv* 2407.01392 (2024). https://arxiv.org/abs/2407.01392.
4. Dhariwal, P. & Nichol, A. Diffusion Models Beat GANs on Image Synthesis. *Advances in Neural Information Processing Systems* (2021). https://arxiv.org/abs/2105.05233.
5. Yu, S. et al. Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think. *International Conference on Learning Representations* (2025). https://arxiv.org/abs/2410.06940.
6. van den Oord, A., Kalchbrenner, N. & Kavukcuoglu, K. Pixel Recurrent Neural Networks. *Proceedings of the 33rd International Conference on Machine Learning* **48**, 1747–1756 (2016). https://proceedings.mlr.press/v48/oord16.html.
7. van den Oord, A. et al. Conditional Image Generation with PixelCNN Decoders. *arXiv* 1606.05328 (2016). https://arxiv.org/abs/1606.05328.
8. Germain, M., Gregor, K., Murray, I. & Larochelle, H. MADE: Masked Autoencoder for Distribution Estimation. *Proceedings of the 32nd International Conference on Machine Learning* **37**, 881–889 (2015). https://arxiv.org/abs/1502.03509.
9. Hu, C., Tai, Y., Vergari, A., Keller, F. & Suglia, A. MIXAR: Scaling Autoregressive Pixel-based Language Models to Multiple Languages and Scripts. *arXiv* 2604.11575v2 (2026). https://arxiv.org/abs/2604.11575v2.
10. Holtzman, A., Buys, J., Du, L., Forbes, M. & Choi, Y. The Curious Case of Neural Text Degeneration. *International Conference on Learning Representations* (2020). https://arxiv.org/abs/1904.09751.
11. See, A., Liu, P. J. & Manning, C. D. Get To The Point: Summarization with Pointer-Generator Networks. *Proceedings of ACL*, 1073–1083 (2017). https://aclanthology.org/P17-1099/.
12. Cui, Y. et al. A Span-Extraction Dataset for Chinese Machine Reading Comprehension. *Proceedings of EMNLP-IJCNLP*, 5883–5889 (2019). https://aclanthology.org/D19-1600/.
13. Zhang, C. et al. Unifying Distributional Training for One-Step Visual Generation. *arXiv* 2609.35763v4 (2026). https://arxiv.org/abs/2609.35763v4.
14. Van Assel, H., De Brouwer, E., Saremi, S., Scalia, G. & Regev, A. Generate in Reconstruction Space, Match in Semantic Space: Transport Geometry for One-Step Generation. *arXiv* 2606.00514v2 (2026). https://arxiv.org/abs/2606.00514v2.
15. Tao, H. et al. Multimodal Flow: Unified Flow Modeling of Language and Vision in Embedding Spaces. *arXiv* 2609.40362v1 (2026). https://arxiv.org/abs/2609.40362v1.
16. Huang, L. & Wu, X. OCG: Optical Character Grounded Visual Reasoning with Flow Matching. Author manuscript (2026). Peer-review status unverified.
17. Bai, S. et al. Qwen3-VL Technical Report. *arXiv* 2511.21631v2 (2025). https://arxiv.org/abs/2511.21631v2.
18. Databricks. Databricks Dolly 15k instruction dataset (2023). https://huggingface.co/datasets/databricks/databricks-dolly-15k.